💬 **Chapter 1**
Continuous-Time and Discrete-Time Signals
- Energy and power
- Signal energy and power

Transformations of the Independent Variable
- Even and Odd Signals

Exponential and Sinusoidal Signals
The Unit Impulse and Unit Step Functions
- Discrete-Time condition
- Continuous-Time condition

Basic System Properties
- With/without memory
- Invertibility
- Causality
- Stability
- Time Invariance
- Linearity

💬 **Chapter 2**
Discrete-Time LTI Systems
- Unit Impulse Response
- Convolution-Sum

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Properties of LTI Systems
Differential or Difference Equations

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- Derivation of DTFT
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💬 Chapter 1

Continuous-Time and Discrete-Time Signals

Energy and power

- **Total energy** over the time interval $t_1 \leq t \leq t_2$ is

$$E_R = \int_{t_1}^{t_2} p(t)\,dt = \int_{t_1}^{t_2} \frac{1}{R} v^2(t)\,dt$$

- **Average power** over the time interval $t_1 \leq t \leq t_2$ is

$$P_R = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2} p(t)\,dt = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2} \frac{1}{R}v^2(t)\,dt$$

Signal energy and power

- Similarly, for any signal $x(t)$ or $x[n]$, the **total energy** is defined as

$$E = \int_{t_1}^{t_2} |x(t)|^2\,dt \quad t_1 \le t \le t_2$$
$$E = \sum_{n=n_1}^{n_2} |x[n]|^2 \quad n_1 \le n \le n_2$$

- Finite-energy signal: $E_\infty < \infty$

$$P_\infty = \lim_{T\to\infty} \frac{E_\infty}{2T} = 0$$
$$P_\infty = \lim_{N\to\infty} \frac{E_\infty}{2N+1} = 0$$

- Finite-power signal: $P_\infty < \infty,\ E_\infty = \infty$
- Infinite energy & power signal: $P_\infty \to \infty,\ E_\infty \to \infty$

Transformations of the Independent Variable

Even and Odd Signals

- Any signal can be broken into a sum of two signals
  - One even and one odd

$$x(t) = x_e(t) + x_o(t)$$
$$x_e(t) = E_v\{x(t)\} = \frac{1}{2}[x(t) + x(-t)]$$
$$x_o(t) = O_d\{x(t)\} = \frac{1}{2}[x(t) - x(-t)]$$

Exponential and Sinusoidal Signals

Continuous-Time Complex Exponential Signals

- General case

$$x(t) = ce^{at} \quad C\ and\ a\ are\ complex\ number$$

- Reax exponential signal
  - C and a are real
  - $a > 0$, as $t \uparrow$, $x(t) \uparrow$
  - $a < 0$, as $t \uparrow$, $x(t) \downarrow$
  - $a = 0$, $x(t)$ is constant

Continuous-Time Complex Sinusoidal Signals

- Sinusoidal Signals $x(t) = Acos(\omega_0 t + \phi)$
- Closely related to complex exponential signals

$$e^{j(\omega_0 t+\phi)} = cos(\omega_0 t + \phi) + jsin(\omega_0 t + \phi)$$
$$Acos(\omega_0 t + \phi) = A \cdot Re\{e^{j(\omega_0 t+\phi)}\}$$
$$Asin(\omega_0 t + \phi) = A \cdot Im\{e^{j(\omega_0 t+\phi)}\}$$

- Fundamental frequency ω_0

$$Acos(\omega_0 t+\phi)=\frac{A}{2}e^{j\phi}e^{j\omega_0 t}+\frac{A}{2}e^{-j\phi}e^{-j\omega_0 t}$$

Sinusoidal signals

- c is real and specifically 1; β is purely imaginary

  $x[n]=e^{j\omega_0 n}$

  Closely related

$$Acos(\omega_0 n+\phi)$$
$$e^{j\omega_0 n}=cos\omega_0 n+jsin\omega_0 n$$
$$Acos(\omega_0 n+\phi)=A/2\cdot e^{j\phi}e^{j\omega_0 n}+A/2\cdot e^{-j\phi}e^{-j\omega_0 n}$$

- Infinite total energy but finite average power $\left|e^{j\omega_0 n}\right|^2=1$

General Signals

$$x[n]=C\alpha^n$$
$$C=|C|e^{j\theta},\alpha=|\alpha|e^{j\omega_0}$$
$$x[n]=|C||\alpha|^n cos(\omega_0 n+\theta)+j\,|C||\alpha|^n sin(\omega_0 n+\theta)$$

The Unit Impulse and Unit Step Functions

Discrete-Time condition

- Unit impulse is defined as

$$\delta[n]=\begin{cases}0, & n\neq 0\\ 1, & n=0\end{cases}$$

- Unit step is define as

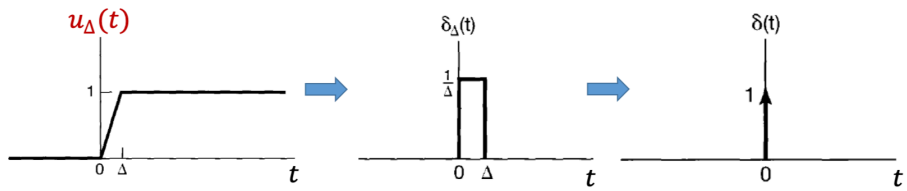
$$u[n]=\begin{cases}0, & n<0\\ 1, & n\geq 0\end{cases}$$

That means, $u[n]=\sum_{m=-\infty}^{n}\delta[m]$ and $\delta[n]=u[n]-u[n-1]$

Let **m = n - k**, we have $u[n]=\sum_{k=\infty}^{0}\delta[n-k]$

Continuous-Time condition

- $u(t)$ is discontinuous at $t=0$, How we get $\delta(t)$?
  - *Consider* $u_\Delta(t)$



$$u(t)=\lim_{\Delta\to 0}u_\Delta(t)\qquad \delta_\Delta(t)=\frac{d\,u_\Delta(t)}{dt}\qquad \delta(t)=\lim_{\Delta\to 0}\delta_\Delta(t)$$

  - arrow at $t=0$: area of the pulse is concentrated at $t=0$
  - arrow height and "1": *area* of the impulse

$u(t)=\int_{-\infty}^{t}\delta(\tau)d\tau$ Let $\sigma=t-\tau$, we have $u(t)=\int_0^\infty\delta(t-\sigma)d\sigma$

Basic System Properties

With/without memory

- With memory:
  - Output depends on the current or previous inputs (or both).
- Without memory:
  - Output depends only on the current memory.

Invertibility

Assume there is a **System A**, put $x[n]$ into it, the output is $y[n]$.

At the same time, there exists a **System B**, when the input is $y[n]$, the output will be $x[n]$.

After that, connect the two systems together, and when we put $x[n]$ into the cascade, the output is still $x[n]$.

Under that circumstance, **System A & B** are inverse systems of each other.

- Noninvertible
  - Obviously, there is a kind of system like:

    $$y[n] = 0$$

    Every input leads to the same $y[n]$, so it's noninvertible.

Causality

- **Definition: The output doesn't depend on the future input.**

Stability

When the $x[n]/x(t)$ is bounded, $i.e.$ $|x[n]| \leq B$ or $|x(t)| \leq B$ for every **n/t**, leads to a bounded output, that is: $|y[n]| \leq B$ or $|y(t)| \leq B$ for every **n/t**

Time Invariance

When there is a time translation on the input signal, for example:

$$x'(t) = x(t - t_0)$$

A time invariance system means the output signal translates at the same time, such as

$$f\{x'(t)\} = f\{x(t - t_0)\} = y(t - t_0)$$

Linearity

- **Definition: A system satisfying the superposition property (additivity and homogeneity)**

$$f\{ax_1(t) + bx_2(t)\} = ay_1(t) + by_2(t)$$

💬 Chapter 2

Discrete-Time LTI Systems

Unit Impulse Response

- **Definition: The system output when the input is $\delta[n]$**
  - For example,

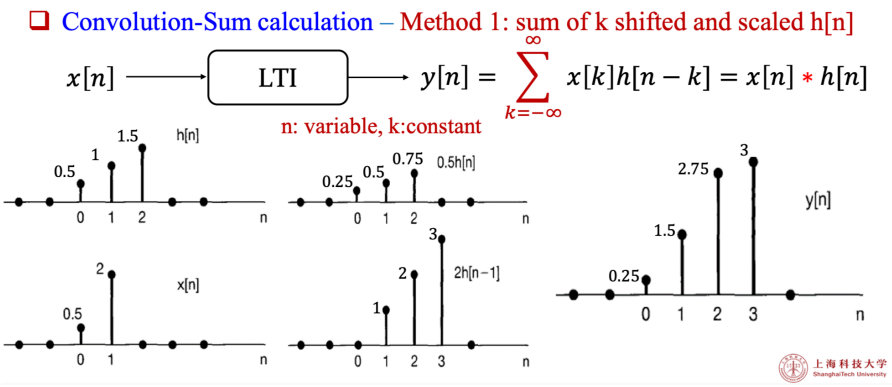
$$y[n] = a_1 x[n] + a_2 x[n-1] + a_3 x[n-2]$$
$$h[n] = a_1 \delta[n] + a_2 \delta[n-1] + a_3 \delta[n-3]$$

Convolution-Sum

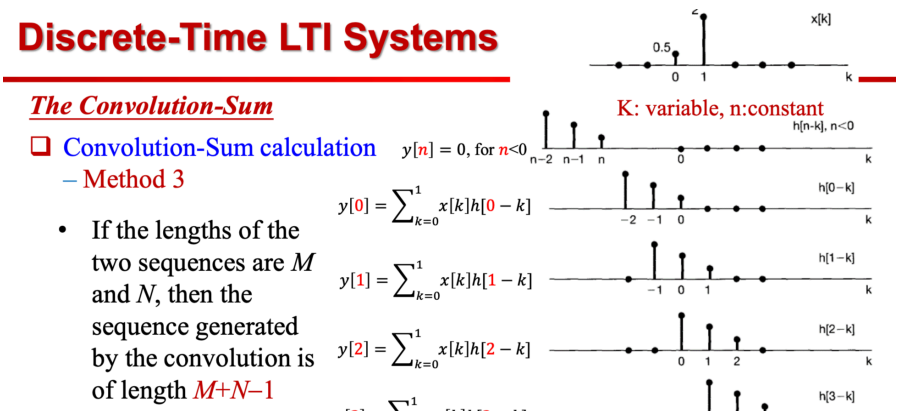
- **Discrete-Time condition**
  - Equation

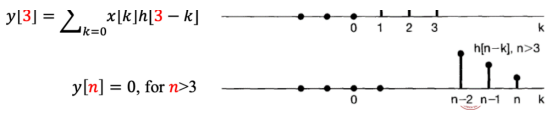
$$x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

  - Method 1:
    - Popularly speaking, a convolution sum can be comprehended by imaging like a kinda superposition of $x[n]$ for each n.
    - Since every impulse, the output responds every time. Take $x[n] = u[n]$ as an example:
    - This input signal means impulse is put into the system every second / every n, when $n \geq 0$. After being taken into the system, it causes the system to response, aka impulse response. Just the way in the graph below.



    - (Add every input "**impulse**" and the ***wave*** it causes together)
  - Method 2:





- For example, $y[1] = \sum_{k=0}^{1} x[k]h[1-k]$.
- Because $h[1-k]$ is an inversion of $h[k]$, so the most right side of its graph represents the very first time that an output wave should be like. (actually it should be like the value of it times the value of the corresponding input value).

Continuous-Time LTI Systems

Properties of LTI Systems

Differential or Difference Equations

💬 Chapter 3

Convergence of the FS

- **Condition 1**
  - If $\int_T |x(t)|^2 \, dt < \infty$, x(t) can be represented by a FS
- **Condition 2 (satisfies 1 & 2 & 3 at the same time)**
  - (1) Absolutely integrable $\int_T |x(t)| \, dt < \infty$
  - (2) In any finite interval of t, $x(t)$ is of ***bounded*** variation; finite maxima and minima in one period
  - (3) $\forall$ ***finite interval of*** t, only a ***finite number*** of ***finite*** discontinuities

Properties of continuous-time FS

Property	Section	Periodic Signal	Fourier Series Coefficients
		$\left.\begin{matrix} x(t) \\ y(t) \end{matrix}\right\}$ Periodic with period T and fundamental frequency $\omega_0 = 2\pi/T$	a_k b_k
Linearity	3.5.1	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Time Shifting	3.5.2	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0} = a_k e^{-jk(2\pi/T)t_0}$
Frequency Shifting		$e^{jM\omega_0 t} = e^{jM(2\pi/T)t} x(t)$	a_{k-M}
Conjugation	3.5.6	$x^*(t)$	a^*_{-k}
Time Reversal	3.5.3	$x(-t)$	a_{-k}
Time Scaling	3.5.4	$x(\alpha t), \alpha > 0$ (periodic with period T/α)	a_k
Periodic Convolution		$\int_T x(\tau)y(t-\tau)d\tau$	$Ta_k b_k$
Multiplication	3.5.5	$x(t)y(t)$	$\sum_{l=-\infty}^{+\infty} a_l b_{k-l}$
Differentiation		$\frac{dx(t)}{dt}$	$jk\omega_0 a_k = jk\frac{2\pi}{T}a_k$
Integration		$\int_{-\infty}^{t} x(t)\,dt$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{jk\omega_0}\right)a_k = \left(\frac{1}{jk(2\pi/T)}\right)a_k$
Conjugate Symmetry for Real Signals	3.5.6	$x(t)$ real	$\begin{cases} a_k = a^*_{-k} \\ \Re e\{a_k\} = \Re e\{a_{-k}\} \\ \Im m\{a_k\} = -\Im m\{a_{-k}\} \\ \lvert a_k\rvert = \lvert a_{-k}\rvert \\ \sphericalangle a_k = -\sphericalangle a_{-k} \end{cases}$
Real and Even Signals	3.5.6	$x(t)$ real and even	a_k real and even

Real and Odd Signals	3.5.6	$x(t)$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals		$\begin{cases} x_e(t) = \mathcal{E}v\{x(t)\} & [x(t) \text{ real}] \\ x_o(t) = \mathcal{O}d\{x(t)\} & [x(t) \text{ real}] \end{cases}$	$\mathcal{R}e\{a_k\}$ $j\mathcal{I}m\{a_k\}$

Parseval's Relation for Periodic Signals

$$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{+\infty} |a_k|^2$$

💬 Chapter 5

DTFT

Derivation

Let $\hat{x}[n]$ be a periodic signal, and $x[n]$ is a period of $\hat{x}[n]$ when $-N_1 \le n \le N_2$, otherwise 0.

So,

$$\hat{x}[n] = \sum_{k\in\langle N\rangle} a_k e^{jk\omega_0 n}$$

Thus, according to DTFS, we have:

$$\begin{aligned} a_k &= \frac{1}{N}\sum_{n=-N_1}^{N_2} \hat{x}[n]e^{-jk(2\pi/N)n} \\ &= \frac{1}{N}\sum_{n=-N_1}^{N_2} x[n]e^{-jk(2\pi/N)n} \\ &= \frac{1}{N}\sum_{n=-\infty}^{\infty} x[n]e^{-jk(2\pi/N)n} \end{aligned}$$

Let $X(e^{j\omega}) = \sum\limits_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$, then we have:

$$a_k = \frac{1}{N}X(e^{jk\omega_0}) \Rightarrow \hat{x}[n] = \frac{1}{N}\sum_{k\in\langle N\rangle} X(e^{jk\omega_0})e^{-jk\omega_0 n}$$

Converting into an integral expression:

$$\begin{aligned} \hat{x}[n] &= \frac{1}{2\pi}\sum_{k\in\langle N\rangle} X(e^{jk\omega_0})e^{-jk\omega_0 n}\cdot\omega_0 \\ &\xlongequal[\hat{x}\to x]{N\to\infty} \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{-j\omega n}\,d\omega \end{aligned}$$

In all, the complete DTFT is

$$\begin{cases} X(e^{j\omega}) = \sum\limits_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \\ x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{-j\omega n}\,d\omega \end{cases}$$

DTFT of signal $x[n] = e^{j\omega_0 n}$: $X(e^{j\omega}) = \sum\limits_{l=-\infty}^{\infty} 2\pi\delta(\omega-\omega_0-2\pi l)$

> 猜出来的，因为CTFT of $x(t) = e^{j\omega_0 t}$ 是类似形式，DTFT应是periodic的

Hence,

$$\begin{cases} x[n] = \sum\limits_{k=\langle N\rangle} a_k e^{jk(2\pi/N)n} = \sum\limits_{k=\langle N\rangle} a_k e^{jk\omega_0 n} \\ X(e^{j\omega}) = \sum\limits_{k=-\infty}^{+\infty} 2\pi a_k\delta(\omega - \frac{2\pi k}{N}) = \sum\limits_{k=-\infty}^{+\infty} 2\pi a_k\delta(\omega - k\omega_0) \end{cases}$$

in which a_k, $\delta\{\cdot\}$ are periodic, $T = N$.

Properties of DTFT

Time shifting & frequency shifting

$$\begin{cases} x[n-n_0] \overset{\mathcal{F}}{\leftrightarrow} e^{j\omega n_0} X(e^{j\omega}) \\ e^{j\omega_0 n} x[n] \overset{\mathcal{F}}{\leftrightarrow} X(e^{j(\omega-\omega_0)}) \end{cases}$$

$X(e^{j(\omega-\omega_0)})$ is frequency shifting to a new frequency which is ω_0 higher than origin. This equals original signal times $e^{j\omega_0 n}$.

- Conjugation property

$$x[n] \overset{\mathcal{F}}{\leftrightarrow} X(e^{j\omega}) \Rightarrow x^*[n] \overset{\mathcal{F}}{\leftrightarrow} X^*(e^{-j\omega})$$

- Conjugation Symmetry

$$X(e^{j\omega}) = X^*(e^{-j\omega})(x[n]\ real) \Rightarrow \begin{cases} \mathcal{E}v\{x[n]\} \overset{\mathcal{F}}{\leftrightarrow} \mathcal{R}e\{X(e^{j\omega})\} \\ \mathcal{O}d\{x[n]\} \overset{\mathcal{F}}{\leftrightarrow} j\mathcal{I}m\{X(e^{j\omega})\} \end{cases}$$

Differencing and accumulation

Time expansion

- Define $x[an]$
  - a should be an integer and $a > 1$
- Definition

$$x_{(k)}[n] = \begin{cases} x[\frac{n}{k}], & n\ is\ a\ multiple\ of\ k \\ 0, & otherwise \end{cases}$$

$$\begin{aligned} X_{(k)}(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x_{(k)}[n] e^{-j\omega n} \\ &\overset{n=rk}{=} \sum_{r=-\infty}^{\infty} x_{(k)}[rk] e^{-j\omega rk} \\ &\overset{x_{(k)}[rk]=x[r]}{=} \sum_{r=-\infty}^{\infty} x[r] e^{-j(k\omega)r} \\ &= X(e^{jk\omega}) \end{aligned}$$

Differentiation in frequency

$$nx[n] \overset{\mathcal{F}}{\leftrightarrow} j\frac{dX(e^{j\omega})}{d\omega}$$

Proof:

$$\begin{aligned} \frac{dX(e^{j\omega})}{d\omega} &= \sum_{n=-\infty}^{\infty} -jnx[n]e^{-j\omega n} \\ \Rightarrow -jnx[n] &\overset{\mathcal{F}}{\leftrightarrow} \frac{dX(e^{j\omega})}{d\omega} \\ \Rightarrow nx[n] &\overset{\mathcal{F}}{\leftrightarrow} j\frac{dX(e^{j\omega})}{d\omega} \end{aligned}$$

Parseval's Relation

反映能量守恒

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{2\pi} |X(e^{j\omega})|^2 \, d\omega$$

Convolution property

💬 Chapter 6

Characterization of Systems

Magnitude & phase

$$\begin{cases} x(t) \overset{\mathcal{F}}{\leftrightarrow} X(j\omega) \\ X(j\omega) = |X(j\omega)|e^{j\angle X(j\omega)} \end{cases}$$

For LTI system:

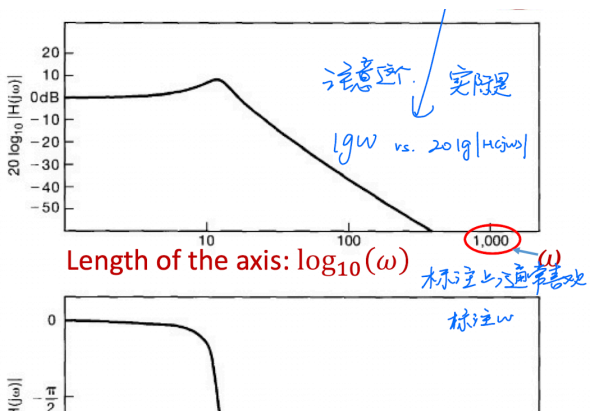
$$x(t) \to h(t) \to y(t) = x(t) * h(t)$$
$$X(j\omega) \to H(j\omega) \to Y(j\omega) = H(j\omega)X(j\omega)$$

Group delay

- Consider a system with $\angle H(j\omega)$ a nonlinear function of ω.
- For a narrow band input $x(t)$, $\angle H(j\omega) \approx -\phi - \alpha\omega$
  - $Y(j\omega) \approx X(j\omega)|H(j\omega)|e^{-j\phi}e^{-j\alpha\omega}$
- The time delay α is referred to as the ***group delay*** at $\omega = \omega_0$.
  - $\tau(\omega) = -\dfrac{d}{d\omega}\{\angle H(j\omega)\}$

Bode Plots

Plots of $20\log_{10}|H(j\omega)|$ and $\angle H(j\omega)$ verses $\log_{10}(\omega)$.



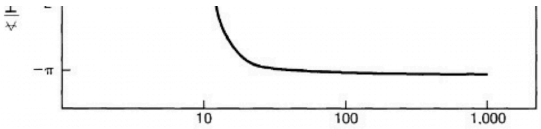


Figure 6.8 A typical Bode plot. (Note that ω is plotted using a logarithmic scale.)